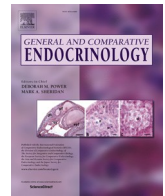




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## General and Comparative Endocrinology

journal homepage: [www.elsevier.com/locate/ygcen](http://www.elsevier.com/locate/ygcen)Explorations in comparative molecular neuroendocrinology: a tribute to Robert M. Does<sup>☆</sup>

## 1. Introduction to special issue

The field of comparative endocrinology has benefited greatly from the scientific contributions and professional leadership of Robert M. Does (Fig. 1). With research spanning over four decades and many taxa, Bob's work has helped establish a comparative framework for understanding the evolution of peptide hormone precursors, particularly those of the opioid/nociceptin family, their post-translational processing and functional products, and the receptors with which they interact to produce complex neuroendocrine signaling systems.

Bob's influence on the field extends well beyond his own research contributions. Throughout his career he has been deeply engaged in our scientific community through mentorship, collaboration, and service. Bob has mentored a great number of undergraduates and graduate students whose research projects frequently led to peer-reviewed publications, and he has collaborated with many laboratories throughout



**Fig. 1.** Robert M. Does, Ph.D. Professor Emeritus at the University of Denver (Denver, CO, USA), Editor-in-Chief of *General and Comparative Endocrinology* from 2001 to 2016, and President of the North American Society for Comparative Endocrinology from 2015 to 2017. Credit: University of Denver.

the world. Notably, he served as Editor-in-Chief of *General and Comparative Endocrinology* (GCE) and President of the North American Society for Comparative Endocrinology (NASCE). Through these efforts he has helped shape not only the direction of comparative endocrinology research, but also the community of scientists who continue to advance the field.

This special issue of GCE serves to recognize Bob's decades of scientific exploration and contributions to comparative endocrinology. From early biochemical studies of neuropeptide processing to current investigations on the functional evolution of melanocortin receptors and their accessory proteins, Bob's work consistently emphasized the importance of studying endocrine systems within a mechanistic, comparative, and functional evolutionary framework. In doing so, Bob's research has helped reveal how peptide hormone systems diversify and functionally evolve while maintaining conserved physiological roles across vertebrate lineages.

The articles assembled here reflect the continuing vitality of the mechanistic, comparative, and evolutionary approaches that Bob has championed throughout his career. Articles in this issue examine the evolution and processing of neuropeptide precursors and the functional diversification of their receptors and accessory proteins across a broad range of taxa, including invertebrates, cyclostomes, teleosts, and tetrapods. Other studies extend these themes to integrative physiological contexts, including stress axis regulation and organismal responses to environmental change. Together, these contributions illustrate the continued integration of molecular, functional, and comparative approaches that have long characterized Bob's work.

## 2. A career in comparative neuroendocrinology

Bob received his B.A. from the College of St. Thomas in 1972, followed by an M.S. from the University of San Francisco in 1974, and a Ph.D. from the University of Minnesota in 1979. Following completion of his graduate studies, Bob pursued postdoctoral training at the University of Colorado School of Medicine and later at the University of Michigan Mental Health Research Institute.

Bob's career in comparative endocrinology has largely tracked the emergence of modern molecular approaches to studying vertebrate neuroendocrine systems. His doctoral and postdoctoral training occurred during a formative period in neuropeptide biology when advances in peptide biochemistry and immunohistochemistry were revealing the diversity of peptide hormones in vertebrate endocrine

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systems. During these years, he worked and published alongside leading investigators in neuroendocrinology, including Huda Akil, Stanley Watson, Richard Mains, and Betty Eipper, whose laboratories were pioneering biochemical, molecular, and neuroanatomical approaches to studying opioid and melanocortin peptide systems. During these early years of training, Bob also worked alongside Aubrey Gorbman and Hiroshi Kawauchi, foundational figures in modern comparative endocrinology. These experiences helped shape Bob's long-standing interest in the biosynthesis, processing, and evolution of peptide hormone precursors through a comparative lens.

Bob joined the Department of Biological Sciences at the University of Denver in 1985, where he remained for the duration of his academic career and built a productive and influential research program in comparative neuroendocrinology. On faculty, he served the department and university in numerous leadership roles, including Associate Chair of the department from 1989 to 1994 and Department Chair from 1994 to 2010. In addition to his administrative leadership, Bob remained deeply committed to undergraduate research, mentoring, and pre-professional advising, including serving as Director of the Undergraduate Research Center and Director of the Pre-professional/Allied Health Advising Center. Although Bob has recently retired, he continues to be active in his scholarship as Emeritus Professor.

Beyond his home institution, Bob has played an important role in the broader comparative endocrinology community through scholarship, collaboration, and leadership that is described in greater detail in the following sections.

### 3. Major scientific contributions

Through a sustained program spanning four decades, Bob has authored or co-authored more than 150 peer-reviewed publications garnering over 5,000 citations. Notably, Bob was first-author on roughly half of these publications, reflecting a career marked not only by broad collaboration but also by sustained intellectual leadership across multiple phases of his own research program and across a transforming technical landscape within the field of comparative neuroendocrinology.

#### 3.1 Opioid peptides and neuropeptide processing

Bob's early research in neuroendocrinology occurred around the time investigators were beginning to recognize that many peptide hormones could be produced through the proteolytic processing of larger precursor proteins. During the late 1970s and early 1980s, advances in peptide biochemistry and neuroendocrine physiology revealed the complexity of these prohormone systems, a concept emerging from pioneering studies in several biomedical laboratories with which Bob was closely associated. Within this rapidly developing biomedical field, Bob's work focused on understanding how these peptide systems were organized and processed across diverse organisms, reflecting an early commitment to the comparative perspective that would define much of his career.

Through numerous biochemical and immunocytochemical studies, Bob and his collaborators investigated opioid peptide systems across a broad phylogenetic range of taxa. These investigations examined the distribution and processing of  $\beta$ -endorphin and related peptides in the pituitary and central nervous system and demonstrated that a single precursor protein could generate multiple biologically active peptides through differential post-translational processing (Dores and Surprenant, 1983; Mains et al., 1983; Dores and Akil, 1987). Working alongside Aubrey Gorbman on a study of Pacific hagfish (*Eptatretus stoutii*) and sea lamprey (*Petromyzon marinus*), Bob demonstrated that Met-enkephalin- and Leu-enkephalin-like peptides are present in the brains of cyclostomes, providing some of the earliest biochemical evidence that the pro-enkephalin opioid system originated early in vertebrate evolution (Dores and Gorbman, 1990).

Bob's work highlighted how proteolytic processing of opioid precursors is a deeply conserved mechanism that varies in its deployment

across tissues and vertebrate lineages. This recognition provided a conceptual framework for understanding how gene duplication, divergence, and variation in prohormone processing could contribute to the evolutionary diversification of neuropeptide systems. These insights would later guide Bob's extensive investigations into the evolution of melanocortins and related peptide families across vertebrates.

#### 3.2 Evolution of melanocortin and related neuropeptide precursors

Among the peptide precursor systems that became central to Bob's research program, proopiomelanocortin (POMC) was especially useful for addressing questions of comparative and evolutionary neurobiology. This multifunctional precursor gives rise to several physiologically important peptides, including adrenocorticotrophic hormone (ACTH), multiple melanocyte-stimulating hormones (MSHs), and  $\beta$ -endorphin. Because these peptides are generated through differential processing of a single precursor protein, the organization of the POMC gene provides a powerful framework for examining how endocrine signaling systems diversified over evolutionary time (Dores and Baron, 2011).

A recurring feature of Bob's work is his ability to recognize evolutionary patterns in neuropeptide systems, often identifying their broader biological implications before the molecular mechanisms were fully understood. An early example of this comes from an immunohistochemical analysis of the brook lamprey (*Lampetra lamoteni*) pituitary, through which Bob revealed ACTH-like immunoreactivity in the rostral pars distalis and MSH-like immunoreactivity in the pars intermedia (Dores et al., 1984). Although these observations relied on antibody cross-reactivity rather than molecular identification, they suggested that melanocortin peptides had already diversified prior to the emergence of this basal vertebrate lineage, a conclusion later supported by the discovery of proopiomelanotropin (POM) and proopiocortin (POC) in lampreys (Takahashi et al., 2005).

Similar evolutionary insight is evident in later molecular studies of basal ray-finned fishes. In characterizing a prepronociceptin (PNOC)-like precursor in white sturgeon (*Acipenser transmontanus*), work from Bob's lab revealed sequence organization and receptor-binding properties of the sturgeon PNOC that were apparently ancestral to the mammalian PNOC ortholog and divergent from the proenkephalin (PENK) and prodynorphin (PDYN) opioid precursors. This work provided evidence that the mammalian PNOC gene represents a derived member of a distinct opioid precursor family (Danielson et al., 2001). PNOC would later be more broadly identified and described as the result of a local duplication of POMC that has been retained in all jawed vertebrates (Sundström et al., 2010).

In both cases, Bob's work demonstrated how a comparative framework could reveal evolutionary relationships among neuropeptide systems long before the genomic organization of these genes was resolved. To Bob, the POMC precursor, with its modular arrangement of melanocortin and opioid peptide domains, proved especially valuable for tracing how complex opioid precursor structure originated and evolved. The breadth of comparative analyses undertaken by Bob and colleagues across basal vertebrates during this phase of his career demonstrated that elements of the melanocortin and opioid peptide systems were present earlier in vertebrate evolution than was appreciated at the time, and that related peptide precursor genes such as POMC, PNOC, PENK, and PDYN diversified from a common ancestral gene. These insights established an evolutionary context for understanding melanocortin signaling and set the stage for the next phase of Bob's research program, which focused on the receptors and accessory proteins that mediate the physiological actions of melanocortin peptides.

#### 3.3 Evolution of melanocortin receptors and accessory proteins

Understanding the evolution of melanocortin signaling requires examining not only the peptide hormones derived from POMC but also the receptors that mediate their physiological effects. Melanocortin

peptides act through a family of G protein-coupled receptors (MC1R–MC5R) that regulate diverse physiological processes including pigmentation, energy balance, and adrenal steroidogenesis (Cone, 2006). Comparative analyses of these receptors across vertebrates have revealed how gene duplication and functional divergence shaped the melanocortin signaling network. Bob contributed significantly to this area by helping to establish an evolutionary framework for understanding the diversification of melanocortin receptors and their functional interactions with the peptides that activate them (Dores et al., 2014).

Particular attention has focused on the melanocortin-2 receptor (MC2R), the receptor for ACTH, which plays a central role in regulating the vertebrate hypothalamic–pituitary–adrenal/interrenal (HPA/HPI) axis. In mammals, MC2R requires interaction with melanocortin receptor accessory proteins (MRAPs) for proper trafficking and signaling, a feature that distinguishes it from other melanocortin receptors (Hinkle and Sebag, 2009; Webb and Clark, 2010). Comparative studies from Bob's laboratory have explored how MC2R, MRAP proteins, and the stress axis co-evolved during vertebrate evolution, highlighting the molecular innovations that accompanied the emergence of the vertebrate stress responses (Dores and Garcia, 2015).

Functional studies in basal vertebrates have provided particularly important evolutionary insights into these molecular relationships. A major advance in understanding the functional evolution of MC2R came from the investigations into the MC2R of a cartilaginous fish, the elephant shark (*Callorhynchus milii*). Work from Bob's laboratory demonstrated that this receptor could be activated in the absence of MRAP (Reinick et al., 2012). This unexpected finding indicated that the strict MRAP dependence observed in mammals represents a derived feature that evolved relatively recently in vertebrate history.

This line of investigation in Bob's laboratory has most recently extended to additional basal vertebrate lineages, including the agnathans, which possess only two ancestral receptor types (MCAR and MCBR). Functional analyses of melanocortin receptors in lamprey further revealed receptor subtypes that lack a strong preference for ACTH over MSH-sized ligands and do not require MRAP for signaling, and provided insight into the organization of the HPI axis in this basal vertebrate lineage (Shaughnessy et al., 2025). Likewise, recent analyses of melanocortin receptors in hagfish revealed similarly promiscuous signaling properties, further suggesting that early melanocortin receptors operated without MRAP dependence and without strong ligand specialization (Shaughnessy et al., 2026).

By integrating peptide evolution with receptor function, Bob's work has demonstrated that comparative structural and functional analyses of hormones and their receptors can explain the emergence and evolution of vertebrate endocrine signaling pathways. His application of this framework within the opioid and melanocortin systems has established a strong reference for future investigators applying similar approaches to other endocrine systems.

#### 4. Mentorship, leadership, and collaboration in comparative endocrinology

While Bob's scientific contributions have significantly advanced our understanding of the evolution of neuropeptide systems, an equally important aspect of Bob's career has been his commitment to mentoring trainees, fostering collaborative scientific exploration, and providing leadership within the global community of comparative endocrinologists.

Bob's mentorship style has been described by colleagues as combining intellectual generosity with genuine enthusiasm for scientific discovery. Over the course of his career, he has trained an extraordinary number of undergraduate and graduate students who have gone on to pursue careers in academia, research institutes, medicine, and industry. Bob has mentored more than 40 graduate students to M.S. and Ph.D. degrees and guided nearly 70 undergraduate researchers through Honors theses, many of whom earned co-authorship on peer-reviewed



Fig. 2. Bob Dores (center) working alongside trainees at the University of Denver. Credit: University of Denver.

publications coming out of his laboratory. Despite this large number of trainees, Bob maintained a remarkably hands-on approach to mentorship, working directly with students at the bench and providing guidance in both experimental design and scientific reasoning (Fig. 2). His influence therefore extends well beyond his own publications, continuing through the work of the scientists he trained and through the professional networks that developed through his laboratory.

Bob's scientific influence is perhaps most apparent through his strong promotion of collaborative research. Throughout his career he has collaborated with nearly 50 independent research groups, fostering an environment in which comparative questions were encouraged and unusual model systems were viewed as opportunities rather than obstacles. Colleagues will attest to his enthusiastic curiosity about organisms studied by others, particularly when those species represented phylogenetically informative lineages relevant to his research interests. This trait was humorously acknowledged during the “Stress Axis” symposium at the 19th International Congress of Comparative Endocrinology (ICCE) in Sendai, Japan, a session organized by Bob and the two authors of the present commentary and held unofficially in Bob's honor. As one speaker remarked during the symposium, Bob had a well-known habit of identifying potential collaborations by way of searching for brain and other tissues from relatively obscure taxa. The comment captured an essential feature of Bob's scientific approach: by examining neuropeptides, receptors, and endocrine signaling systems across an exceptionally wide range of organisms, he consistently used collaboration and comparative approaches to uncover evolutionary relationships that might have otherwise remained unrecognized.

Bob has also played an important leadership role in shaping the global comparative endocrinology community. In the late 2000s he was instrumental in helping establish NASCE as an independent scientific society and biennial meeting following its separation from the Society for Integrative and Comparative Biology. Bob later served as President of NASCE from 2015 to 2017, a tenure that culminated in a highly successful joint meeting with the 18th ICCE in Lake Louise, Canada. NASCE continues to provide a central forum for organizing the comparative endocrinology community in North America while fostering communication and collaboration with comparative endocrinology societies and researchers around the world.

Bob's leadership within our community is also evidenced by his involvement with *GCE*. He has contributed greatly to the success of this journal, where he has served on the Editorial Board for nearly three decades and as Editor-in-Chief from 2001 to 2016. During this period, he guided the journal through a time when molecular and genomic approaches were rapidly expanding the scope of comparative

endocrinology. At the same time, he ensured that the journal continued to represent comparative research across a wide diversity of organisms well beyond traditional biomedical models. As Editor-in-Chief of *GCE*, Bob was instrumental in building strong connections between the journal and the broader comparative endocrinology community, including helping formalize a close relationship between *GCE* and NASCE. He also initiated the journal's series of special issues, creating a venue for highlighting emerging areas of research and recognizing influential scientists whose work has shaped the field. The present tribute issue is itself a continuation of that tradition. Bob continues to serve on the Editorial Board of *GCE*.

Through his mentorship, collaborations, and scientific leadership, Bob has left a lasting impact on the field of comparative endocrinology. His career is reflected not only in his influential publications but also in the community of scientists he helped train, inspire, and connect.

## 5. Legacy and future directions in comparative endocrinology

The body of work described above reflects a scientific career guided by a consistent intellectual principle: that understanding endocrine systems requires examining them across the full diversity of invertebrate and vertebrate life. Through studies of neuropeptide precursors, receptor evolution, and endocrine signaling pathways, Bob has exemplified this principle, demonstrating the importance of a comparative approach to understanding the evolutionary processes that give way to complex physiological systems. His research repeatedly showed that insights into vertebrate endocrinology often emerge most clearly when examined in organisms that occupy key phylogenetic positions.

Today, comparative endocrinology is entering a period of rapid expansion driven by readily available genomic and transcriptomic sequencing, highly accurate predictions of protein structure, and functional assays that can more easily be applied to non-traditional model organisms. These approaches now allow investigators to test many of the evolutionary hypotheses that Bob and his collaborators first articulated using biochemical, immunological, and early molecular techniques. As increasing numbers of vertebrate genomes become available and functional studies expand to additional lineages, the comparative framework that guided Bob's research will remain essential for interpreting the evolutionary origins of endocrine signaling systems.

Equally important is the scientific culture that Bob helped foster within our field. His career illustrates how curiosity, mentorship, collaboration, and leadership can shape both scientific discovery and the communities that support it. As new generations of researchers enter endocrinology through a comparative lens, the community that Bob helped build will undoubtedly continue to play an important role in their scientific development and professional fulfillment.

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